



BCX53 series

80 V, 1 A PNP medium power transistors

Rev. 11 — 12 August 2026

Product data sheet

1. General description

PNP medium power transistors in a SOT89 (SC-62) flat lead Surface-Mounted Device (SMD) plastic package.

2. Features and benefits

- High current
- Three current gain selections
- High power dissipation capability
- Exposed heatsink for excellent thermal and electrical conductivity
- AEC-Q101 qualified^[1]

[1] Part of the products (as listed in the [revision history](#)) have removed automotive qualification status.

3. Applications

- Linear voltage regulators
- High-side switches
- Battery-driven devices
- Power management
- MOSFET drivers
- Amplifiers

4. Quick reference data

Table 1. Quick reference data

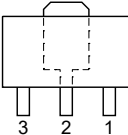
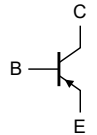
$T_{amb} = 25\text{ °C}$ unless otherwise specified.

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
V_{CEO}	collector-emitter voltage	open base		-	-	-80	V
I_C	collector current			-	-	-1	A
I_{CM}	peak collector current	single pulse; $t_p \leq 1\text{ ms}$		-	-	-2	A
h_{FE}	DC current gain						
	BCX53	$V_{CE} = -2\text{ V}$; $I_C = -150\text{ mA}$ $T_{amb} = 25\text{ °C}$	[1]	63	-	250	
	BCX53-10		[1]	63	-	160	
	BCX53-16		[1]	100	-	250	

[1] pulsed; $t_p \leq 300\text{ }\mu\text{s}$; $\delta \leq 0.02$

5. Pinning information

Table 2. Pinning

Pin	Symbol	Description	Simplified outline	Graphic symbol
1	E	emitter		 006aaa231
2	C	collector		
3	B	base		

6. Ordering information

Table 3. Ordering information

Type number	Package		
	Name	Description	Version
BCX53	SOT89	plastic, surface-mounted package; 3 leads; 1.5 mm pitch; 4.5 mm x 2.5 mm x 1.5 mm body	SOT89
BCX53-10			
BCX53-16			

7. Marking

Table 4. Marking

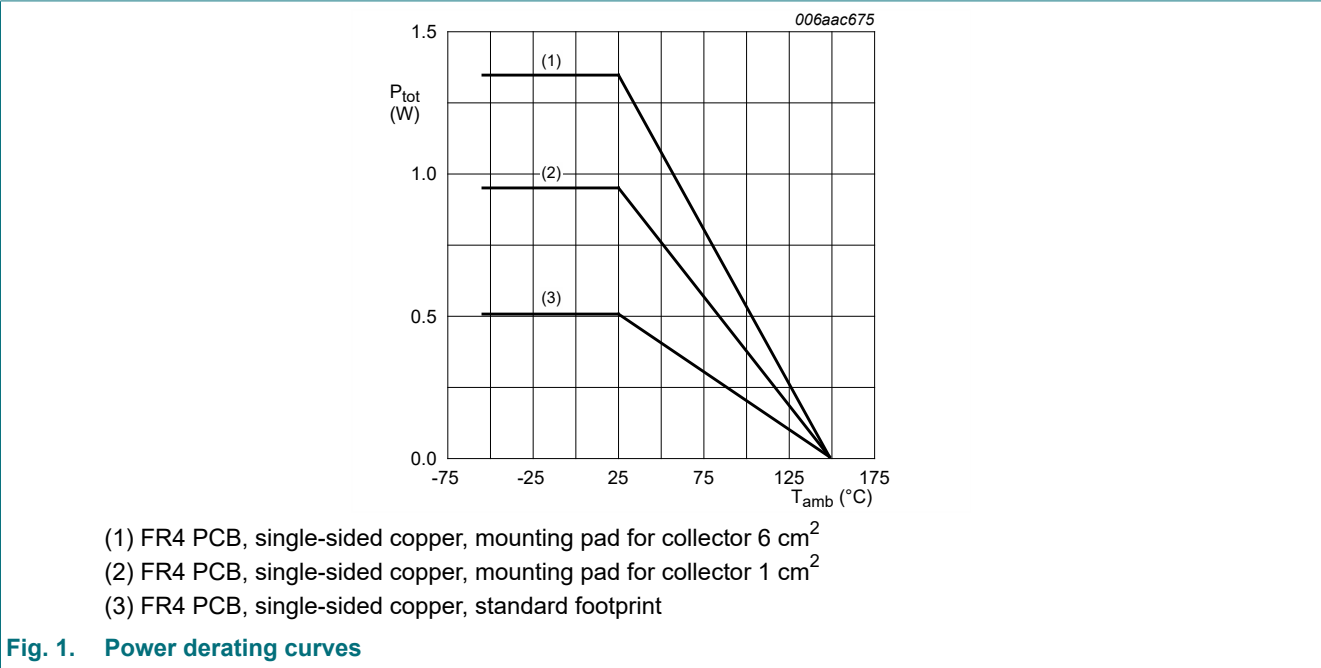
Type number	Marking code
BCX53	AH
BCX53-10	AK
BCX53-16	AL

8. Limiting values

Table 5. Limiting values
In accordance with the Absolute Maximum Rating System (IEC 60134).
 $T_{amb} = 25\text{ }^{\circ}\text{C}$ unless otherwise specified.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CBO}	collector-base voltage	open emitter	-	-100	V
V_{CEO}	collector-emitter voltage	open base	-	-80	V
V_{EBO}	emitter-base voltage	open collector	-	-5	V
I_C	collector current		-	-1	A
I_{CM}	peak collector current	single pulse; $t_p \leq 1\text{ ms}$	-	-2	A
I_B	base current		-	-0.3	A
I_{BM}	peak base current	single pulse; $t_p \leq 1\text{ ms}$	-	-0.3	A
P_{tot}	total power dissipation	$T_{amb} \leq 25\text{ }^{\circ}\text{C}$ [1]	-	0.50	W
		[2]	-	0.95	W
		[3]	-	1.35	W
T_j	junction temperature		-	150	$^{\circ}\text{C}$
T_{amb}	ambient temperature		-55	150	$^{\circ}\text{C}$
T_{stg}	storage temperature		-65	150	$^{\circ}\text{C}$

- [1] Device mounted on an FR4 Printed-Circuit-Board (PCB); single-sided copper; tin-plated and standard footprint.
[2] Device mounted on an FR4 PCB; single-sided copper; tin-plated; mounting pad for collector 1 cm^2 .
[3] Device mounted on an FR4 PCB; single-sided copper; tin-plated; mounting pad for collector 6 cm^2 .



9. Thermal characteristics

Table 6. Thermal characteristics

$T_{amb} = 25\text{ }^{\circ}\text{C}$ unless otherwise specified.

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
$R_{th(j-a)}$	thermal resistance from junction to ambient	in free air	[1]	-	-	250	K/W
			[2]	-	-	132	K/W
			[3]	-	-	93	K/W
$R_{(j-sp)}$	thermal resistance from junction to solder point			-	-	16	K/W

- [1] Device mounted on an FR4 PCB; single-sided copper; tin-plated and standard footprint.
[2] Device mounted on an FR4 PCB; single-sided copper; tin-plated; mounting pad for collector 1 cm².
[3] Device mounted on an FR4 PCB; single-sided copper; tin-plated; mounting pad for collector 6 cm².

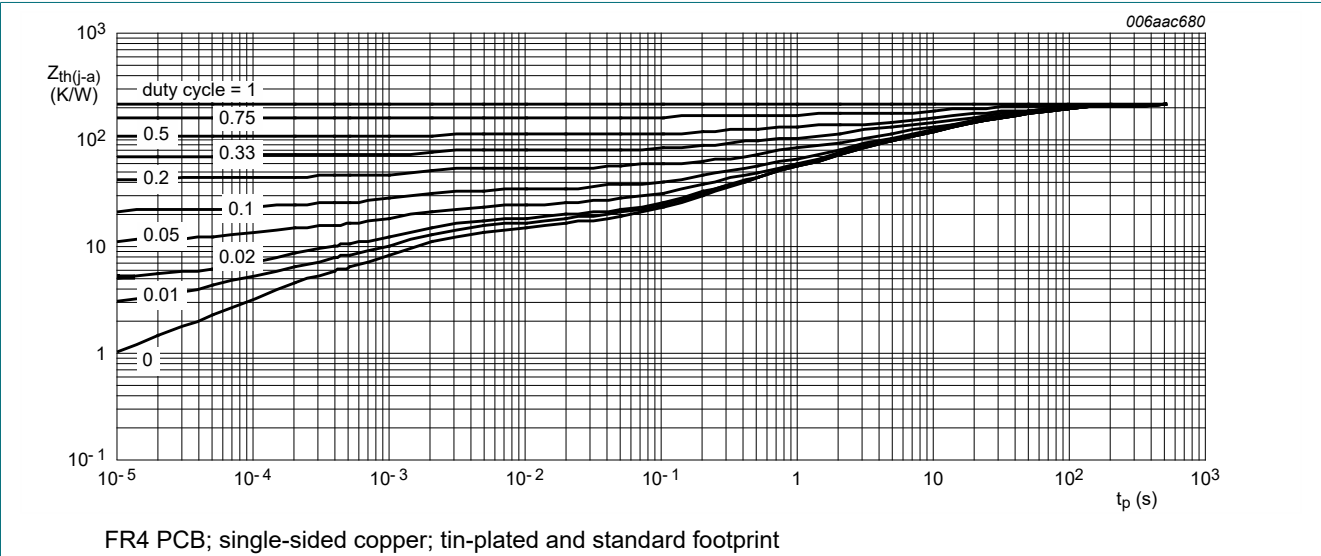


Fig. 2. Transient thermal impedance from junction to ambient as a function of pulse duration; typical values

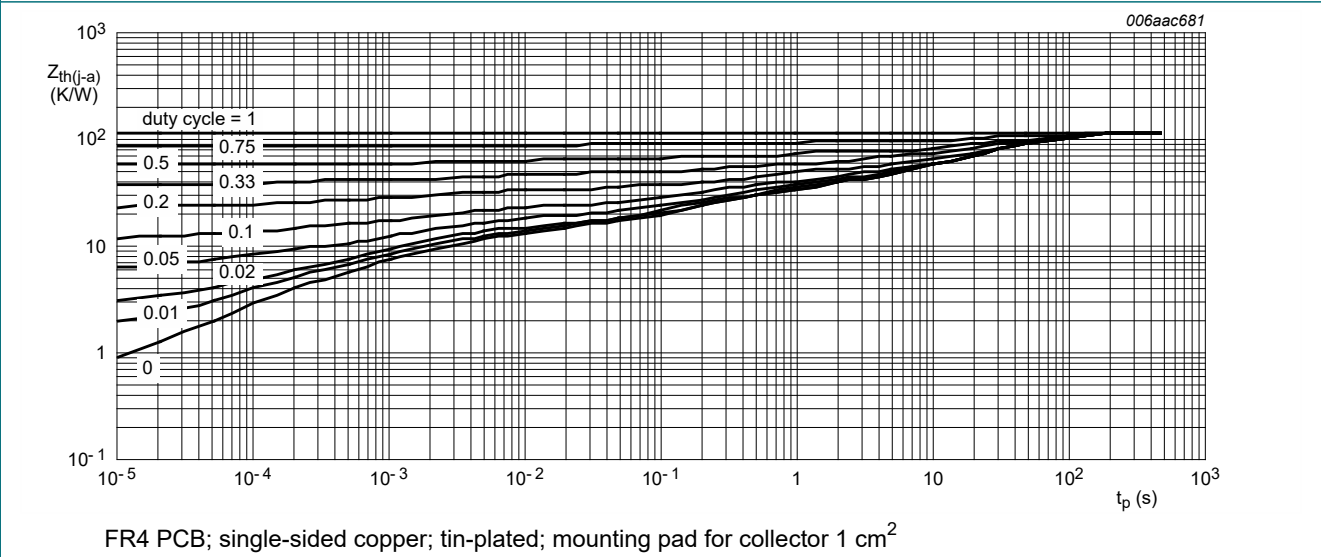
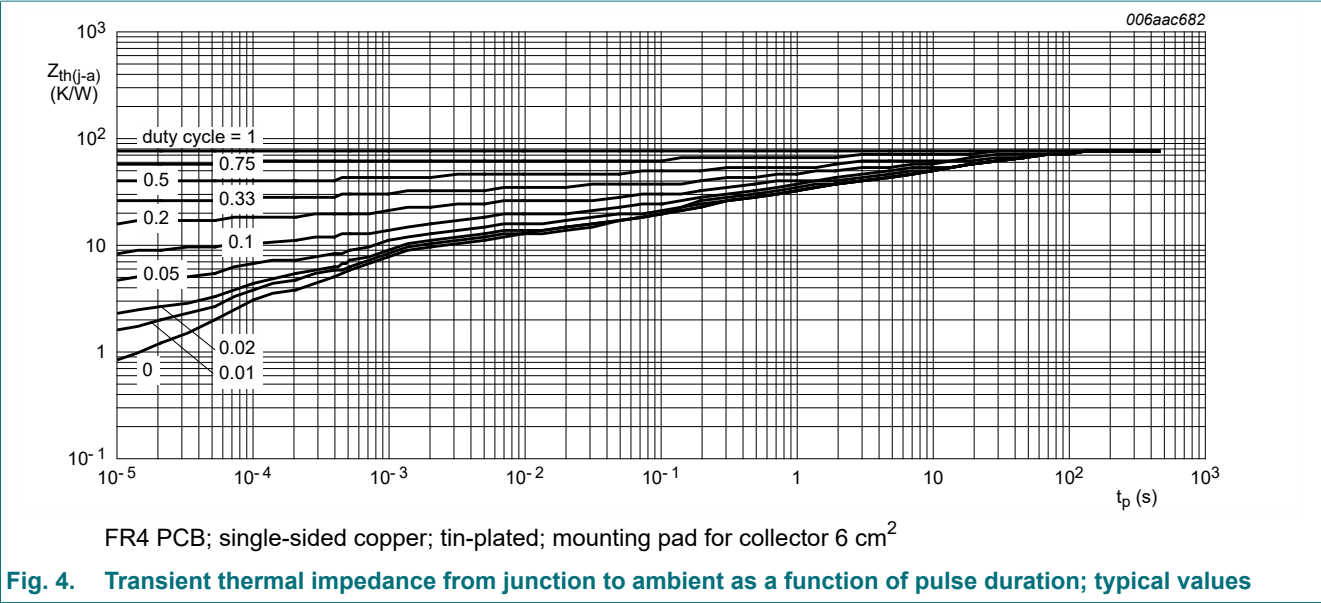


Fig. 3. Transient thermal impedance from junction to ambient as a function of pulse duration; typical values



10. Characteristics

Table 7. Characteristics

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
I_{CBO}	collector-base cut-off current	$V_{CB} = -30\text{ V}; I_E = 0\text{ A}$ $T_{amb} = 25\text{ °C}$		-	-	-100	nA
		$V_{CB} = -30\text{ V}; I_E = 0\text{ A}; T_j = 150\text{ °C}$		-	-	-10	μA
I_{EBO}	emitter-base cut-off current	$V_{EB} = -5\text{ V}; I_C = 0\text{ A}$ $T_{amb} = 25\text{ °C}$		-	-	-100	nA
h_{FE}	DC current gain						
	BCX53	$V_{CE} = -2\text{ V}; I_C = -5\text{ mA}$ $T_{amb} = 25\text{ °C}$	[1]	63	-	-	
		$V_{CE} = -2\text{ V}; I_C = -150\text{ mA}$ $T_{amb} = 25\text{ °C}$		63	-	250	
		$V_{CE} = -2\text{ V}; I_C = -500\text{ mA}$ $T_{amb} = 25\text{ °C}$		40	-	-	
	BCX53-10	$V_{CE} = -2\text{ V}; I_C = -5\text{ mA}$ $T_{amb} = 25\text{ °C}$	[1]	63	-	-	
		$V_{CE} = -2\text{ V}; I_C = -150\text{ mA}$ $T_{amb} = 25\text{ °C}$		63	-	160	
		$V_{CE} = -2\text{ V}; I_C = -500\text{ mA}$ $T_{amb} = 25\text{ °C}$		40	-	-	
	BCX53-16	$V_{CE} = -2\text{ V}; I_C = -5\text{ mA}$ $T_{amb} = 25\text{ °C}$	[1]	63	-	-	
		$V_{CE} = -2\text{ V}; I_C = -150\text{ mA}$ $T_{amb} = 25\text{ °C}$		100	-	250	
		$V_{CE} = -2\text{ V}; I_C = -500\text{ mA}$ $T_{amb} = 25\text{ °C}$		40	-	-	
V_{CEsat}	collector-emitter saturation voltage	$I_C = -500\text{ mA}; I_B = -50\text{ mA}$ $T_{amb} = 25\text{ °C}$	[1]	-	-	-0.5	V
V_{BE}	base-emitter voltage	$V_{CE} = -2\text{ V}; I_C = -500\text{ mA}$ $T_{amb} = 25\text{ °C}$	[1]	-	-	-1	V
C_c	collector capacitance	$V_{CB} = -10\text{ V}; I_E = I_C = 0\text{ A}; f = 1\text{ MHz}$ $T_{amb} = 25\text{ °C}$		-	15	-	pF
f_T	transition frequency	$V_{CE} = -5\text{ V}; I_C = -50\text{ mA}; f = 100\text{ MHz}$ $T_{amb} = 25\text{ °C}$		-	145	-	MHz

[1] pulsed; $t_p \leq 300\text{ }\mu\text{s}$; $\delta \leq 0.02$

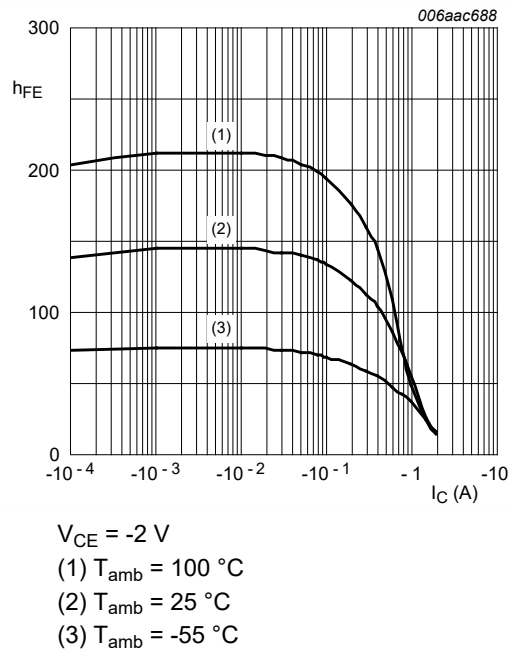


Fig. 5. DC current gain as a function of collector current; typical values

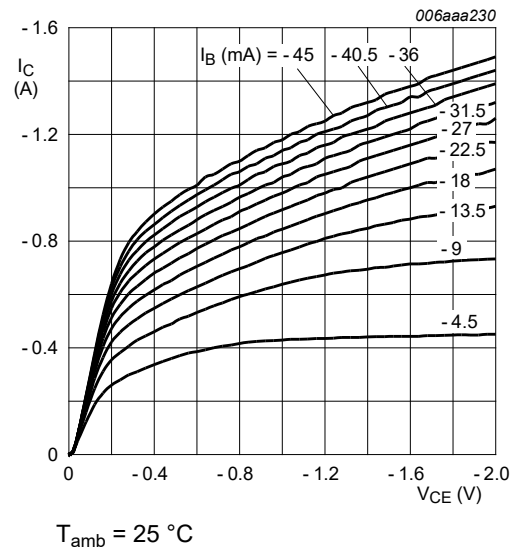


Fig. 6. Collector current as a function of collector-emitter voltage; typical values

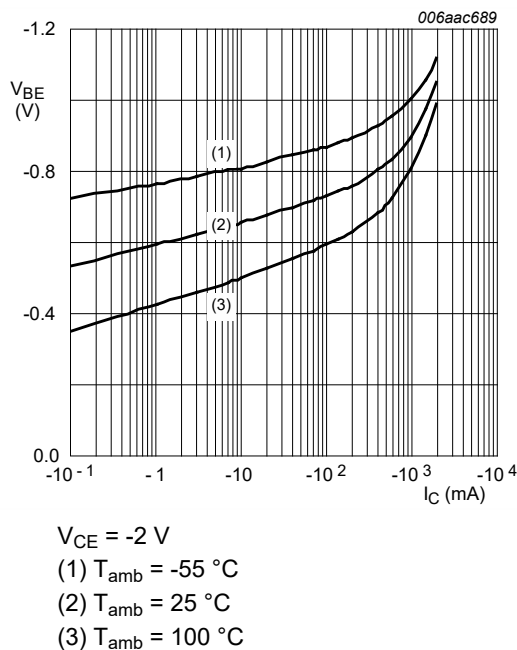


Fig. 7. Base-emitter voltage as a function of collector current; typical values

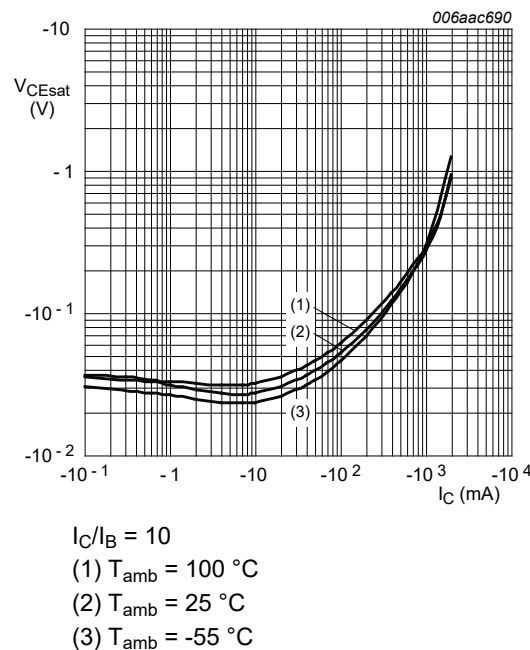


Fig. 8. Collector-emitter saturation voltage as a function of collector current; typical values

13. Soldering

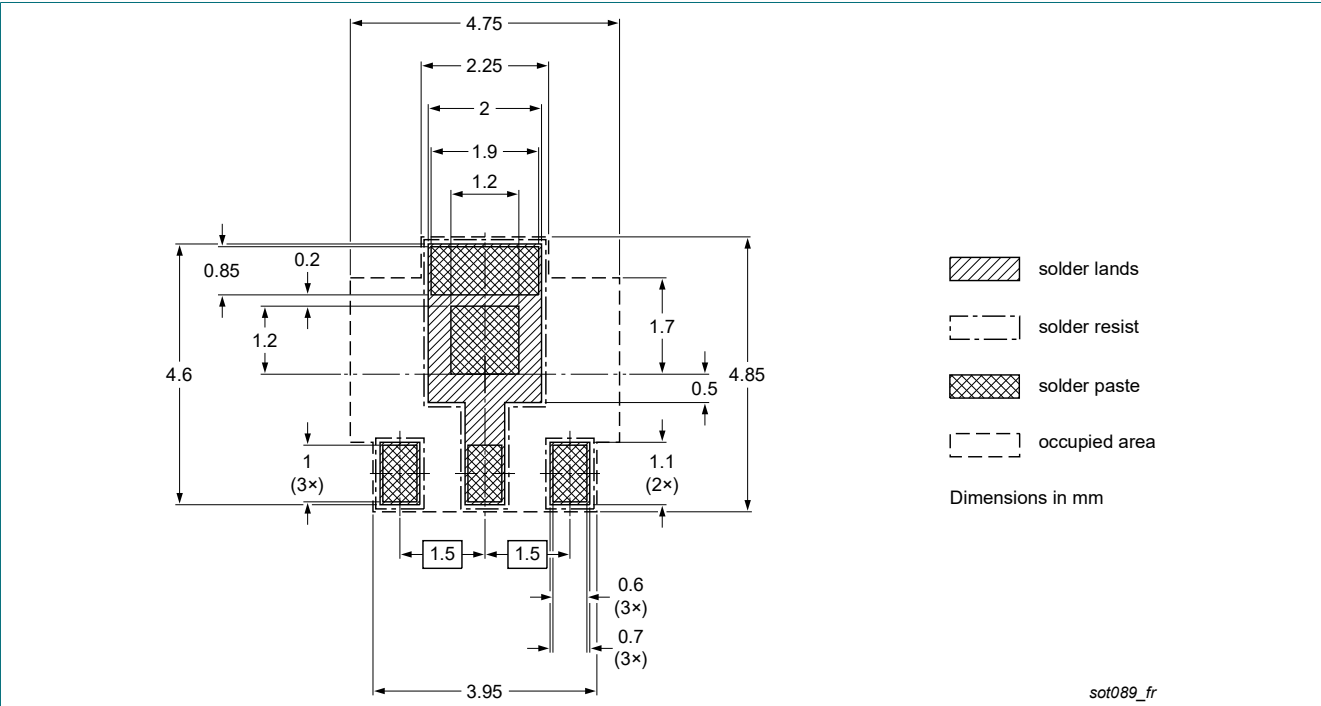


Fig. 10. Reflow soldering footprint for SOT89

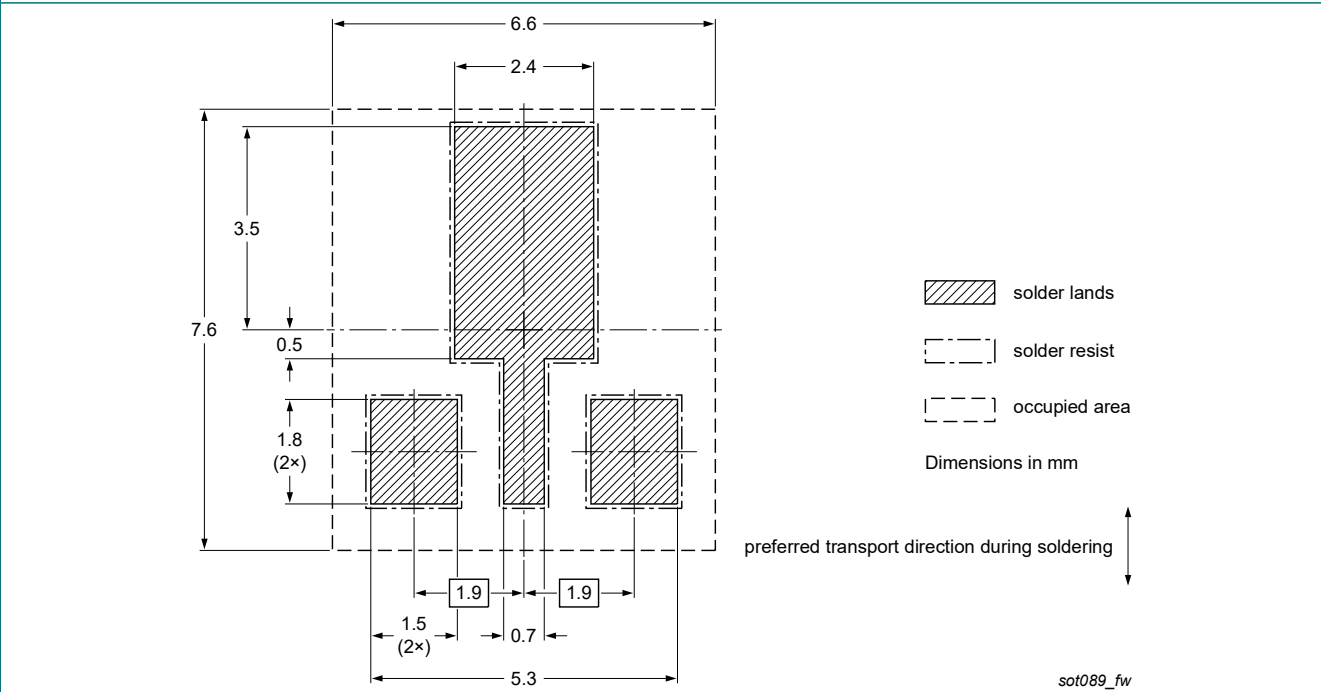


Fig. 11. Wave soldering footprint for SOT89

14. Revision history

Table 8. Revision history

Document ID	Release date	Data sheet status	Change notice	Supersedes
BCX53_SER v.11	20260812	Product data sheet	-	BCX53_SER v.10
Modifications:	Only BCX53 changed to non-automotive qualification. Please refer to nexperia.com for automotive (-Q) product alternative.			
BCX53_SER v.10	20230804	Product data sheet	-	BCP53_BCX53_BC53PA v.9
BCP53_BCX53_BC53PA v.9	20220106	Product data sheet	-	BC640_BCP53_BCX53 v.8
BC640_BCP53_BCX53 v.8	20111021	Product data sheet	-	BC640_BCP53_BCX53 v.7
BC640_BCP53_BCX53 v.7	20070604	Product data sheet	-	BC640_BCP53_BCX53 v.6
BC640_BCP53_BCX53 v.6	20050225	Product data sheet	CPCN200405029	BC636_638_640 v.5 BCP51_52_53 v.5 BCX51_52_53 v.4
BC636_638_640 v.5	20011010	Product specification	-	BCX51_52_53 v.5
BCX51_52_53 v.5	20030206	Product specification	-	BCX51_52_53 v.4
BCX51_52_53 v.4	20011010	Product specification	-	BCX54_55_56 v.3

15. Legal information

Data sheet status

Document status [1][2]	Product status [3]	Definition
Objective [short] data sheet	Development	This document contains data from the objective specification for product development.
Preliminary [short] data sheet	Qualification	This document contains data from the preliminary specification.
Product [short] data sheet	Production	This document contains the product specification.

- [1] Please consult the most recently issued document before initiating or completing a design.
- [2] The term 'short data sheet' is explained in section "Definitions".
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